

Speech-to-Text and Voice Agent Benchmark

A Comprehensive Market Evaluation for a Real-Time AI Receptionist

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Automotive AI Receptionist

Abstract

Real-time conversational AI systems depend on a sequence of latency-sensitive components. Speech-to-Text (STT) represents the first major processing stage after a customer speaks and directly influences downstream language-model reasoning, entity extraction, tool execution, and Text-to-Speech (TTS) generation.

This paper presents a comprehensive benchmark of leading STT and voice-AI providers for an automotive AI receptionist workload. The evaluated providers include Deepgram, AssemblyAI, ElevenLabs, Speechmatics, and Gladia. The evaluation focuses on conventional Word Error Rate (WER), semantic transcription accuracy, business-critical entity accuracy, Time-to-First-Segment (TTFS), total latency, and processing reliability.

The benchmark used a proprietary receptionist dataset consisting of realistic, multi-sentence customer interactions involving service appointments, vehicle information, names, dates, times, and other structured business information.

The results demonstrate different strengths across the evaluated providers. ElevenLabs achieved the lowest measured WER at 0.52%, while Deepgram led in Entity Accuracy at 90.0%. Gladia produced the lowest measured adjusted TTFS at 1216 ms. AssemblyAI and Speechmatics also demonstrated competitive latency.

The study does not establish a universal provider winner. Instead, it provides a quantitative framework for evaluating STT systems according to the accuracy, latency, reliability, and structured-data requirements of a production AI receptionist.

The paper also evaluates the current TTS architecture by comparing Amazon Polly through Twilio, Amazon Polly directly through AWS, and Fish Audio, including usage-based pricing and architectural trade-offs.

1. Introduction

A production AI receptionist must understand natural spoken language accurately while maintaining sufficiently low latency to support natural conversation.

The general voice-agent architecture can be represented as:

Caller → Telephony → STT → LLM / Agent → RAG / Tools → TTS → Caller

The STT component is particularly important because errors introduced during transcription propagate into subsequent stages of the system.

For example, incorrectly recognizing a vehicle model, customer name, appointment date, or appointment time can result in:

- Incorrect CRM lookups
- Incorrect appointment scheduling
- Incorrect customer information
- Additional clarification turns
- Increased conversation latency

At the same time, transcription accuracy cannot be considered independently from latency. A highly accurate STT system that takes several seconds to produce a usable transcript may negatively affect the conversational experience.

The objective of this benchmark is therefore to evaluate the available technologies across both dimensions while considering the specific requirements of an automotive AI receptionist.

2. Evaluation Objectives

The benchmark was designed to evaluate the following characteristics:

1. Conventional transcription accuracy
2. Semantic preservation of customer intent
3. Accuracy of business-critical entities
4. Streaming responsiveness
5. Complete utterance processing latency
6. Processing reliability
7. Suitability for real-time conversational AI

The evaluation also considers the practical implications of integrating each provider into a larger voice-agent architecture.

3. Benchmark Dataset

A proprietary receptionist dataset was constructed to represent realistic automotive customer-service conversations.

The dataset contains 20 relatively long, multi-sentence audio samples per evaluated provider covering scenarios such as:

- Booking service appointments
- Requesting vehicle service
- Providing vehicle makes and models
- Providing model years
- Providing customer names
- Spelling names
- Providing appointment dates
- Providing appointment times
- Updating customer information
- Conversational and rambling speech

The dataset was intentionally designed around information that would have downstream business value rather than only measuring isolated word-level transcription.

4. Evaluation Metrics

Five primary metrics were considered.

4.1 Word Error Rate

Word Error Rate (WER) is the conventional metric for measuring transcription accuracy.

It measures the number of substitutions, deletions, and insertions required to transform the generated transcript into the reference transcript.

Lower WER indicates better conventional transcription accuracy.

4.2 Semantic WER

Semantic Word Error Rate measures whether the intended meaning of a spoken utterance is preserved.

This is important for conversational AI because not every textual difference has the same practical impact.

For example, a minor spelling variation that does not change the meaning of a vehicle model may be less significant than an incorrect appointment time.

Lower Semantic WER indicates better preservation of the customer's intended meaning.

4.3 Entity Accuracy

Entity Accuracy measures the percentage of critical entities correctly recognized.

Examples include:

- Customer names
- Vehicle makes
- Vehicle models
- Model years
- Dates
- Appointment times
- Locations
- Numbers
- Alphanumeric information

This metric is particularly important for an AI receptionist because these entities are frequently passed into CRM, scheduling, and other business systems.

4.4 Adjusted TTFS

Adjusted Time-to-First-Segment (TTFS) measures how quickly usable streaming transcription becomes available.

The reported latency values were adjusted according to the benchmark methodology to account for the initial delay associated with the benchmark setup.

Lower TTFS indicates faster availability of transcription for downstream processing.

4.5 Total Latency

Total latency represents the time required to process the complete utterance.

This metric provides additional context beyond the first available transcription segment and is useful for evaluating complete turn processing.

5. Aggregate Market Results

5.1 Evaluation Snapshot

Testing date: August 17, 2026

The benchmark was executed using the following provider models:

- Deepgram: **nova-3**
- AssemblyAI: **u3-rt-pro**
- ElevenLabs: **scribe_v1**
- Speechmatics: **Standard**
- Gladia: **Solaria-3**

All benchmark results reported in this paper represent testing performed as of **August 17, 2026** using the configurations listed above.

The benchmark evaluated 20 audio samples per provider using the same reference transcripts and evaluation methodology.

The aggregate benchmark results are summarized in Table 1.

Table 1: Aggregate STT and Voice-AI Benchmark Results (tested as of August 17, 2026)

Provider	Model	WER	Entity Accuracy	Adjusted TTFS	Total Latency	Success
Deepgram	nova-3	0.92%	90.0%	1296 ms	14699 ms	100%
AssemblyAI	u3-rt-pro	0.81%	83.8%	1258 ms	14697 ms	100%
ElevenLabs	scribe_v1	0.52%	88.8%	1544 ms	15234 ms	100%
Speechmatics	Standard	2.53%	82.5%	1257 ms	14696 ms	100%
Gladia	Solaria-3	8.26%	83.8%	1216 ms	14697 ms	100%

The aggregate results demonstrate that no single provider leads across every metric.

ElevenLabs produced the lowest measured WER and Semantic WER.

Deepgram produced the highest Entity Accuracy.

Gladia produced the lowest adjusted TTFS.

Speechmatics and AssemblyAI also demonstrated competitive latency.

All providers achieved a 100% success rate across the benchmark samples.

6. Benchmark Visualizations

The following figures visualize the measured differences between providers across transcription accuracy, semantic accuracy, entity recognition, adjusted latency, and total latency.

6.1 Word Error Rate

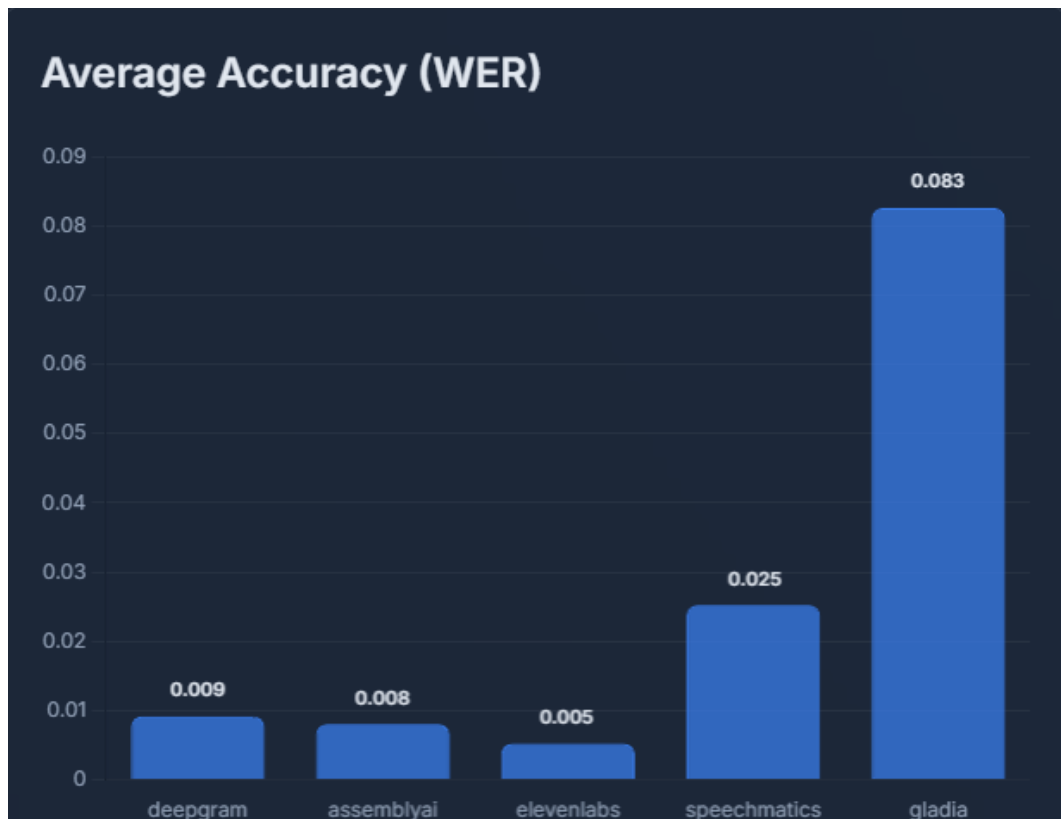


Figure 1: Average Word Error Rate (WER) across evaluated providers. Lower values indicate better transcription accuracy.

The WER results show that ElevenLabs produced the lowest measured error rate, followed by AssemblyAI and Deepgram. Speechmatics and Gladia produced higher measured error rates.

6.2 Semantic WER

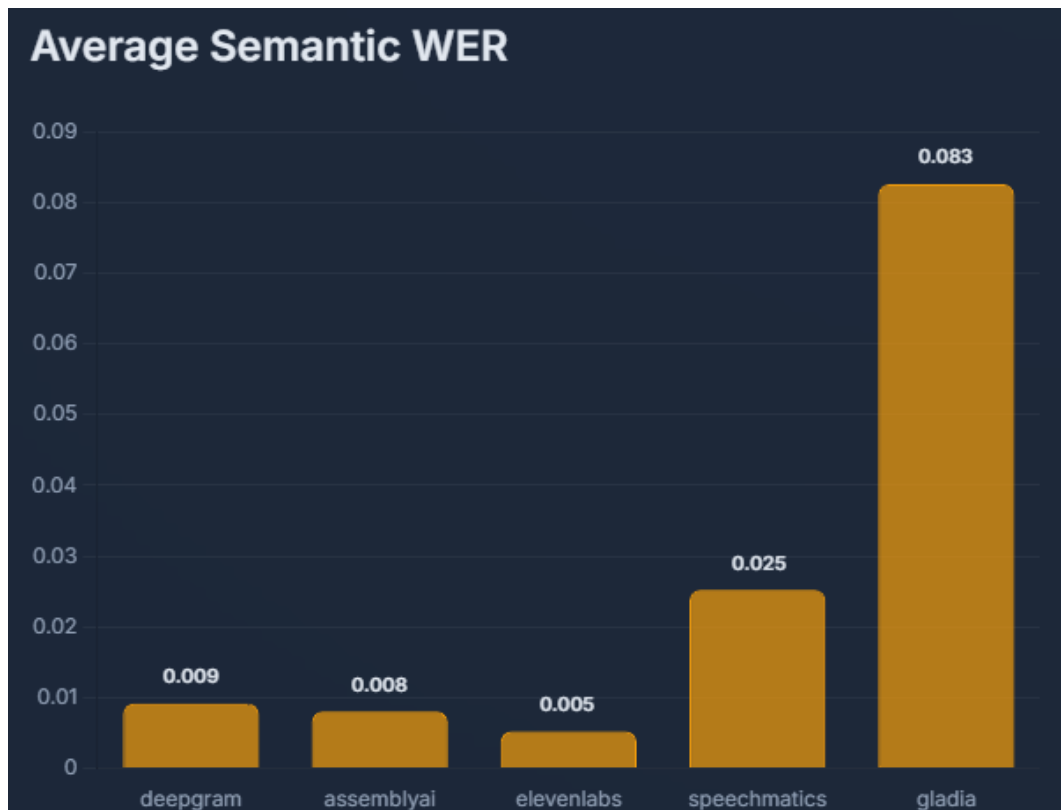


Figure 2: Average Semantic WER across evaluated providers. Lower values indicate better preservation of the intended meaning.

The Semantic WER distribution follows the same general pattern as the conventional WER measurements in the benchmark.

6.3 Entity Accuracy

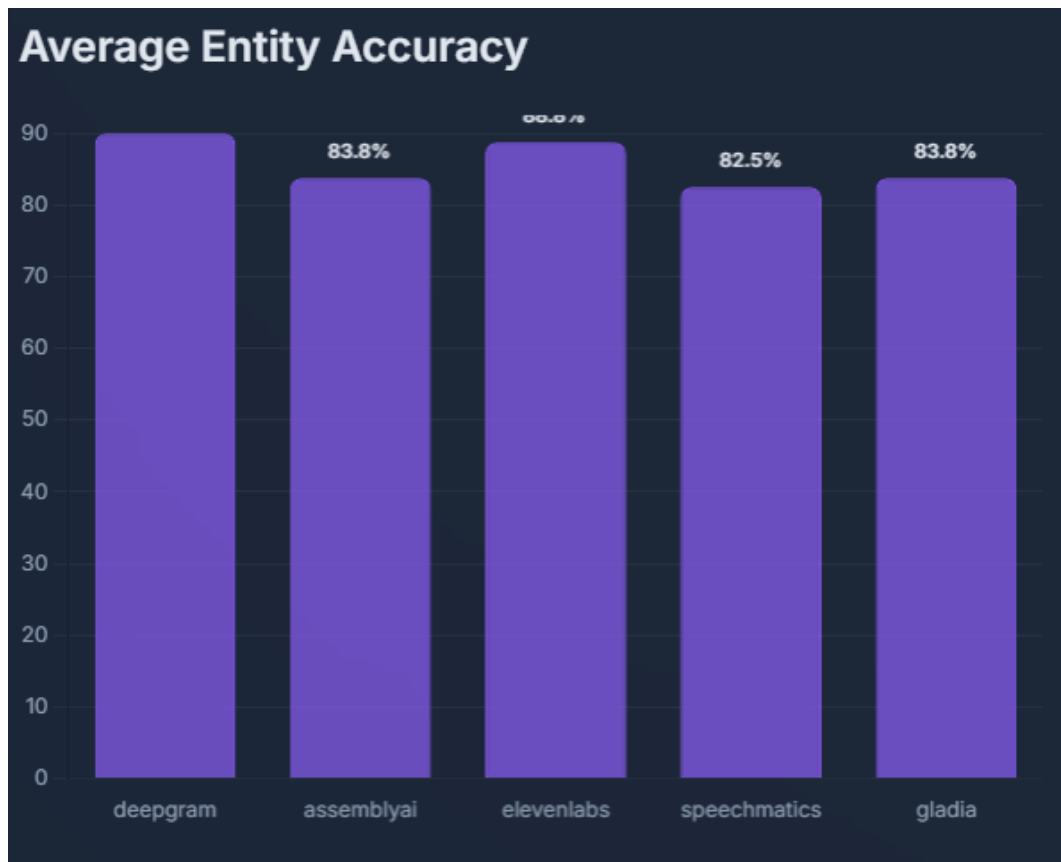


Figure 3: Average Entity Accuracy across evaluated providers. Higher values indicate more accurate recognition of business-critical entities.

Deepgram achieved the highest Entity Accuracy at 90.0%. ElevenLabs followed closely at 88.8%, while AssemblyAI and Gladia achieved 83.8%. Speechmatics recorded 82.5%.

6.4 Adjusted TTFS

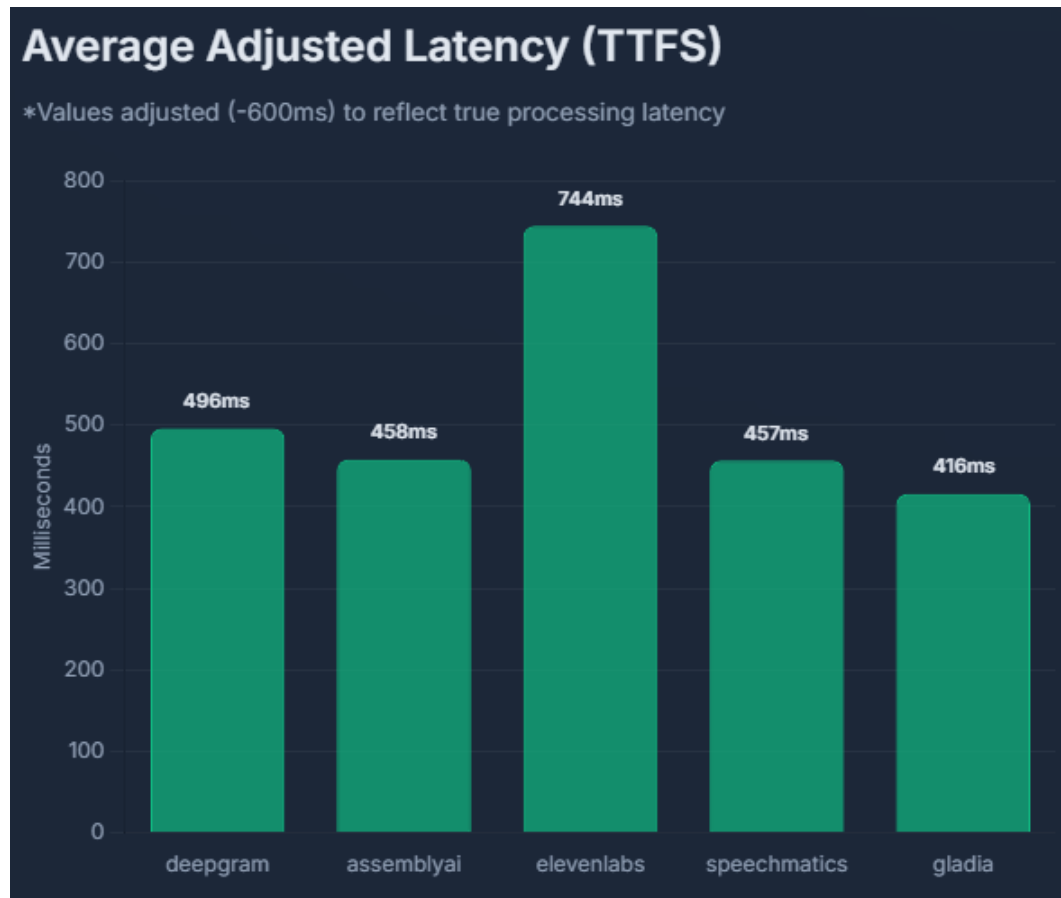


Figure 4: Average adjusted Time-to-First-Segment (TTFS). Lower values indicate faster availability of usable transcription.

Gladia recorded the lowest adjusted TTFS at approximately 1216 ms. Speechmatics and AssemblyAI followed closely at approximately 1257 ms and 1258 ms respectively. Deepgram measured approximately 1296 ms, while ElevenLabs measured approximately 1544 ms.

6.5 Total Latency



Figure 5: Average total processing latency across evaluated providers.

The total latency measurements are relatively close for Deepgram, AssemblyAI, Speechmatics, and Gladia. ElevenLabs recorded the highest total latency in the benchmark.

7. Provider Analysis

7.1 Deepgram

Deepgram, using **nova-3**, demonstrated the strongest entity-recognition performance in the benchmark, achieving an Entity Accuracy of 90.0%.

Its measured WER and Semantic WER were approximately 0.92%.

Deepgram’s adjusted TTFS was approximately 1296 ms.

The entity-accuracy result is particularly relevant to the target application because automotive receptionists frequently process structured information such as vehicle models, model years, customer names, and appointment times.

7.2 AssemblyAI

AssemblyAI, using **u3-rt-pro**, achieved a measured WER and Semantic WER of approximately 0.81%, with an Entity Accuracy of 83.8%.

Its adjusted TTFS was approximately 1258 ms.

During testing, a 20/20 processing success rate was achieved after implementing connection pooling and session-management changes to accommodate concurrency behavior.

7.3 ElevenLabs

ElevenLabs, using **scribe_v1**, produced the lowest measured WER and Semantic WER at approximately 0.52%.

Its Entity Accuracy was approximately 88.8%.

Its adjusted TTFS was approximately 1544 ms.

Its total latency was also the highest at approximately 15234 ms.

These results demonstrate a trade-off between transcription accuracy and streaming responsiveness within the evaluated workload.

7.4 Speechmatics

Speechmatics, using the **Standard** model configuration, achieved a measured WER and Semantic WER of approximately 2.53%.

Its Entity Accuracy was approximately 82.5%.

The adjusted TTFS was approximately 1257 ms.

The results demonstrate competitive streaming latency alongside a higher measured transcription error rate than Deepgram, AssemblyAI, and ElevenLabs in this benchmark.

7.5 Gladia

Gladia, using **Solaria-3**, produced the lowest adjusted TTFS at approximately 1216 ms.

Its measured WER and Semantic WER were approximately 8.26%, while Entity Accuracy was approximately 83.8%.

The benchmark also recorded a 100% processing success rate.

8. Accuracy and Comprehension

The benchmark demonstrates that conventional WER and business-oriented Entity Accuracy provide different views of STT quality.

ElevenLabs achieved the lowest measured WER at approximately 0.52%.

Deepgram, however, achieved the highest Entity Accuracy at approximately 90.0%.

This distinction is important for an AI receptionist.

Consider a customer saying:

“I have a twenty twenty-two Lexus RX and I’d like to schedule service next Tuesday.”

A transcription system may achieve a low conventional WER while still making an error involving the vehicle model or appointment date.

For downstream automation, the entity error may be substantially more consequential than a minor transcription difference elsewhere in the sentence.

Therefore, the benchmark treats Entity Accuracy as a complementary metric rather than relying exclusively on WER.

9. Latency and Real-Time Responsiveness

Latency strongly influences perceived conversational quality.

The measured adjusted TTFS values were:

- Gladia: approximately 1216 ms
- Speechmatics: approximately 1257 ms
- AssemblyAI: approximately 1258 ms
- Deepgram: approximately 1296 ms
- ElevenLabs: approximately 1544 ms

These measurements represent only the STT component of the complete voice-agent pipeline. The system must still perform:

STT → LLM reasoning → Tool/RAG execution → TTS generation → Audio playback

Consequently, an STT system that consumes a significant portion of the latency budget can constrain the remaining stages.

At the same time, optimizing STT latency at the expense of accuracy can result in additional conversational turns, which can ultimately increase total interaction time.

10. Reliability

All evaluated providers achieved a 100% processing success rate.

The benchmark successfully processed 20 samples per provider under the tested conditions.

Reliability should nevertheless be evaluated further under production-like conditions, including:

- Concurrent calls
- Long-running conversations
- Network instability
- Connection interruptions
- Provider rate limits
- Streaming reconnection
- Extended audio sessions

11. Architectural Considerations

The STT engine is only one component of the overall voice-agent architecture.

A production system can be represented as:

Caller → Twilio Voice → Streaming STT → Agent / LLM → RAG / Tools → Streaming TTS → Caller

This architecture introduces several independent latency sources.

For example, an STT system with low TTFS may still provide a slow overall response if LLM reasoning, tool execution, or TTS generation introduces additional latency.

Conversely, an extremely fast STT system may not provide a better user experience if downstream reasoning or TTS introduces additional latency.

Therefore, the final evaluation should consider end-to-end response time rather than optimizing STT in isolation.

12. TTS Evaluation

The voice-agent experience is strongly influenced by the TTS engine.

The current architecture uses Twilio’s Voice platform with Amazon Polly voices through the Twilio <Say> interface.

Three TTS approaches were considered:

1. Amazon Polly through Twilio
2. Amazon Polly directly through AWS
3. Fish Audio

It is important to note that Amazon Polly through Twilio and Amazon Polly Direct use the **same underlying Amazon Polly TTS service**. They are not different TTS providers.

The difference is the integration path and the specific configuration used.

12.1 TTS Test Configuration

TTS test date: August 18, 2026

The project configuration used the following TTS setups:

- **Twilio + Amazon Polly:** Amazon Polly Standard, Joanna voice, 8000 Hz
- **Amazon Polly Direct:** Amazon Polly Neural, Joanna voice, 24000 Hz
- **Fish Audio:** fish-audio-v1, Alex voice, 44100 Hz

The Twilio and direct-AWS configurations therefore use the same Amazon Polly service but different engines and audio configurations.

12.2 TTS Cost Comparison

Pricing snapshot: August 14, 2026

The pricing figures in this section represent the published provider rates used for the comparison as of the stated pricing snapshot date.

Pricing is usage-based and may vary depending on provider, configuration, region, volume, account agreement, and applicable discounts. These values should therefore be interpreted as a point-in-time pricing comparison.

Table 2: TTS Provider and Pricing Comparison (pricing checked as of August 14, 2026)

Configuration	Model / Voice	Published Price	Approx. Cost
Twilio + Amazon Polly	Polly Standard / Joanna	\$0.0008 / 100 characters	\$8 / 1M characters
Twilio + Amazon Polly	Polly Generative	\$0.0032 / 100 characters	\$32 / 1M characters
Amazon Polly Direct	Polly Neural / Joanna	\$16 / 1M characters	\$16 / 1M characters
Amazon Polly Direct	Polly Generative	\$30 / 1M characters	\$30 / 1M characters
Fish Audio	fish-audio-v1 / Alex	\$15 / 1M UTF-8 bytes	\$15 / 1M bytes

Amazon Polly’s direct published pricing is \$4 per million characters for Standard voices, \$16 per million characters for Neural voices, and \$30 per million characters for Generative voices. The direct benchmark configuration uses the Neural engine, therefore the applicable direct Polly benchmark rate is \$16 per million characters. The Generative rate is included as an additional reference configuration.

Twilio provides Amazon Polly Generative voices through its Text-to-Speech integration at a published rate of \$0.0032 per 100 characters, equivalent to approximately \$32 per 1 million characters. Twilio’s TTS pricing is separate from its underlying voice-call charges; telephony usage charges are additional.

Amazon Polly through Twilio and Amazon Polly Direct use the same underlying Amazon Polly technology. The difference is the integration path and the applicable pricing.

Fish Audio’s API pricing is based on input text measured in millions of UTF-8 bytes..

12.3 Standardized Voice Test

The TTS samples were generated on **August 18, 2026** using the configurations described above.

The same receptionist-oriented sentence was used to provide a consistent qualitative comparison:

“Hello, this is the automotive service team. I can help you schedule an appointment for your 2022 Lexus RX. I have availability next Tuesday at 2 PM. Would that work for you?”

The sentence contains a business context, vehicle model, model year, appointment date, appointment time, and a conversational question.

12.4 Generated TTS Voice Samples

The following samples were generated during the TTS evaluation.

Table 3: Generated TTS Voice Samples (tested on August 18, 2026)

TTS Engine	Voice / Configuration	Audio
Amazon Polly(generative)	Matthew	Listen
Fish Audio	ALLE	Listen

The Amazon Polly sample represents the Amazon Polly voice technology evaluated through the tested configurations. The Twilio + Amazon Polly and Amazon Polly Direct configurations use

the same underlying Amazon Polly service while differing in engine, audio configuration, and integration path.

The audio links should point to the final public HTTPS locations of the generated MP3 files.

12.5 Voice Quality Evaluation

The generated samples can be evaluated using the following criteria:

- Naturalness
- Pronunciation
- Conversational pacing
- Pausing
- Emotional expression
- Proper-name pronunciation
- Vehicle-model pronunciation
- Numeric information
- SSML and prosody behavior
- Time-to-first-audio

The audio samples provide a qualitative complement to the quantitative cost and latency analysis and are not, by themselves, sufficient to establish a final TTS provider selection.

12.6 TTS Architecture Trade-offs

Twilio + Amazon Polly:

Provides an integrated telephony and TTS workflow. The tested configuration uses Amazon Polly Standard with the Joanna voice at 8000 Hz, reflecting the telephony-oriented configuration used by the application.

Amazon Polly Direct:

Provides direct access to Amazon Polly through the AWS API. The tested configuration uses the Neural engine with the Joanna voice at 24000 Hz, providing a higher-fidelity audio configuration than the Twilio telephony setup.

Fish Audio:

Provides an independent TTS platform. The evaluated configuration uses the `fish-audio-v1` engine with the Alex voice at 44100 Hz.

The comparison should consider voice quality, latency, SSML support, reliability, integration complexity, audio quality, and total cost rather than TTS generation price alone.

13. TTS Pricing Analysis

The pricing snapshot used for this analysis is:

- **Pricing date:** August 14, 2026
- **Twilio Amazon Polly Standard:** \$0.0008 per 100 characters
- **Amazon Polly Neural:** \$16 per 1 million characters
- **Fish Audio:** \$15 per 1 million UTF-8 bytes

The Twilio figure represents the TTS component of the service. Twilio voice-call charges are separate and should be included when calculating the complete cost of a production call.

For direct Amazon Polly usage, the applicable price depends on the selected Polly engine. The benchmark configuration uses the Neural engine.

For Fish Audio, usage is measured according to the provider's text-input billing unit.

The pricing comparison should therefore be interpreted as a TTS generation cost comparison rather than a complete voice-call cost comparison.

14. Operational Considerations

Beyond raw benchmark performance, production deployment requires consideration of several operational factors.

- API availability
- Rate limits
- Concurrent request capacity
- Streaming stability
- Error handling
- Provider failover
- SDK maturity
- Integration complexity
- Monitoring
- Cost predictability

A multi-provider architecture may also be considered to reduce dependency on a single STT or TTS provider.

For example, a primary STT provider could be paired with a secondary provider for failover, while TTS could similarly support multiple voice-generation paths.

15. Future Evaluation

The benchmark should be extended before making a final infrastructure decision.

Future STT evaluation should include:

- Larger datasets
- Different accents
- Background noise
- Telephone-quality audio
- Different speaking rates
- Concurrent calls
- Long conversations
- Streaming reconnection
- Cost per minute
- Entity-specific accuracy

Future TTS evaluation should include:

- Human naturalness ratings
- Voice quality
- Time-to-first-audio
- Streaming latency
- Prosody
- SSML support
- Interruptibility
- Voice consistency
- Cost per conversation

The complete voice-agent system should ultimately be evaluated using:

- End-of-speech to first-audio latency
- STT latency

- LLM latency
- RAG latency
- Tool execution latency
- TTS latency
- Total response latency
- Barge-in performance
- Call completion rate
- Recovery from provider failures
- Total cost per conversation

16. Open Evaluation

The benchmark results demonstrate that the evaluated providers exhibit different strengths.

ElevenLabs produced the lowest measured WER and Semantic WER.

Deepgram produced the highest Entity Accuracy.

AssemblyAI demonstrated competitive accuracy, latency, and processing reliability.

Gladia produced the lowest measured adjusted TTFS.

Speechmatics demonstrated competitive latency with a higher measured transcription error than several other evaluated providers.

The TTS comparison also presents different trade-offs between integrated telephony, direct provider access, voice configuration, audio quality, latency, operational complexity, and cost.

These results do not establish a final provider selection.

Instead, they provide a quantitative foundation for the next phase of evaluation.

The final infrastructure decision should be based on a complete end-to-end benchmark combining:

**STT Accuracy + Entity Accuracy + STT Latency + LLM Latency + Tool
Latency + TTS Latency + Voice Quality + Reliability + Cost**

The next stage should therefore focus on measuring the complete voice-agent pipeline under realistic production conditions and determining which combination of providers provides the best overall balance for the intended workload.

References

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